Reinforcement Learning (RL) Basics – Cheatsheet

1. Agent and Environment

- **Agent:** robot/software which **performs actions**.
- **Environment:** medium where agent acts and **learns through rewards** until it reaches the final state.

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2. Policy

- **Policy:** rule or instruction that tells the agent **which action to perform in a state**.
- Example: Agent in state S → **policy tells move Right** because that leads to **high-value state**.

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3. Value Function (V)

- **V(s):** total reward agent gets **from this state to goal** following a policy.
- Calculated using **Bellman equation**.
- Example: Neighboring V-values: Up = 3, Right = 5, Down = 2 → move Right.

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4. Q-Function (Q)

- **Q(s,a):** total reward if agent **takes action a from state s** and then follows a policy.
- Calculated using **Bellman equation**.
- Example: Q(S, Up) = 3, Q(S, Right) = 5, Q(S, Down) = 2 → choose action with highest Q → Right.
- **Q-values care about both state and action** → more precise guidance.

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5. Episode

- A **sequence of actions from start to goal**.
- Shows the **path taken**, optimal or not.

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6. Shorter Path

- Shorter path is better because it gives **reward sooner**.
- With **discount factor γ**, rewards received earlier are more valuable.

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7. Bellman Equation

- Helps **calculate value of each state or state-action**.
- High-value states → closer to goal or more reward.
- Policy uses these values to **guide the agent along the optimal path**.

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8. Exploration vs Exploitation

- **Exploration:** agent tries **new actions** to discover rewards.
- **Exploitation:** agent uses **known best path/actions** to reach the goal.

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9. Discount Factor (γ)

- Tells **how much agent cares about future reward**.
- γ close to 1 → agent cares about future, moves slowly state by state.
- γ close to 0 → agent ignores future rewards, focuses on immediate reward.

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10. On-Policy vs Off-Policy

- **On-policy:** agent **learns from actions it actually takes**.
- **Off-policy:** agent **learns about best actions/policy while exploring differently**.

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11. Model-Based vs Model-Free

- **Model-Based:** agent **knows or learns environment** → can **plan actions**.
- **Model-Free:** agent **does not know environment**, learns **by trying actions and seeing rewards**.

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12. Reward Function

- Tells agent **how good or bad an action is** in a state.
- Helps agent **learn best actions** and reach goal efficiently.

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13. Deterministic vs Stochastic Policy

- **Deterministic:** always does **same action in a state**.
  - Works if goal is along straight line, no obstacles.
  - Can fail if obstacles or goal not in straight line.
- **Stochastic:** chooses actions **based on probabilities**.
  - High probability → action more likely.
  - Helps explore paths, avoid obstacles, and reach goal.

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14. Summary of V vs Q vs Policy

- **V-values:** value of **states** → tells which **state is best**.
- **Q-values:** value of **action + state** → tells **which action to take**.
- **Policy:** uses **V or Q-values** to decide the **next action** toward the goal.
- Usually, **both V and Q lead to the same next state**, but Q is more precise because it **considers actions directly**.